

Probabilistic Dataset Merge

Scenario 3: Scale Mismatch

What happens when one feature uses different units across the two datasets

What this scenario demonstrates. One of the most dangerous data errors: the same variable measured in different units in the two datasets. Here the target reports Dragon Sightings as a per-thousand figure (2.469, meaning approximately 2,469), while every row in the supplemental dataset uses the raw count (2,469–7,389). After standardization, the target’s Dragon Sightings value becomes an extreme outlier — pulling every candidate’s distance upward by roughly the same large amount and effectively neutralising that feature’s discriminating power. The system still selects the nearest row, but inspecting the per-feature contribution table alongside the large absolute distance reveals the problem: one feature is responsible for nearly all of the distance. Concentration alone is not a warning — any scenario where only one feature differs will show 100% contribution from that feature. The concern here is that concentration is paired with a large absolute distance, suggesting the dominant feature is not slightly off but fundamentally misscaled.

The Target Record

This is the record we are trying to match. The system will search the supplemental dataset for the row closest to these values.

Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units
2	40.2	99.5	649

Table 1: Target row values (raw, unstandardized).

***Note:** Dragon Sightings in the target is entered as **2.469** (per-thousand units). The supplemental dataset records the same variable as a raw count; the correct raw value for this record is **2,469**. All other features use consistent units across both datasets.*

All Candidate Matches

The table below shows all 20 supplemental rows, sorted from closest to farthest. Distances are computed in standardized space (see Section 3). The highlighted row is the best match — the one the system selects.

Rank	Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units	Distance
1	2,469	40.2	99.5	649	1.4830
2	2,524	30.0	99.5	557	1.9652
3	2,948	33.4	98.5	687	2.0183
4	3,849	42.2	99.5	1,043	2.5782
5	3,885	37.2	100.0	1,004	2.5790
6	2,512	25.8	96.1	521	2.8991
7	4,158	29.8	99.4	1,089	3.0575
8	4,555	26.5	99.4	930	3.2901
9	4,822	25.7	98.2	914	3.5245
10	4,891	34.4	99.6	1,339	3.5960
11	5,094	27.3	100.0	1,041	3.6136
12	5,006	30.1	97.5	1,230	3.7761
13	5,296	25.4	99.3	1,128	3.8890
14	4,223	36.3	94.2	1,132	3.9812
15	4,701	58.7	100.0	1,519	4.3611
16	3,465	47.9	92.5	1,080	4.4105
17	6,303	29.8	100.0	1,490	4.6503
18	6,516	23.3	99.3	1,385	4.8770
19	6,668	32.0	98.5	1,641	5.0156
20	7,389	35.0	100.0	1,851	5.6328

Note: “Distance” is the standardized Euclidean distance between the target and each candidate row. A distance of 0.0000 means a perfect, identical match.

How Distance Is Calculated: A Worked Example

We use the **rank 2** candidate below to illustrate the calculation step by step. (The rank 1 match has distance 0 and is trivial; rank 2 shows the full arithmetic.)

Step 1 — Put every feature on the same scale (standardization).

Raw values are measured in different units (counts, years, percentages). Before computing distance we convert each to a *standardized score*: how many standard deviations above or below the combined average that value sits.

$$z = \frac{\text{raw value} - \text{combined average}}{\text{combined spread}}$$

Feature	Target (raw)	Candidate (raw)	Combined Avg.	Combined Spread
Dragon Sightings	2	2,524	4346.50	1663.25
Avg. Wizard Age	40.2	30.0	33.87	8.34
Pct. Leprechaun Cottages	99.5	99.5	98.60	1.96
Goblin Family Units	649	557	1089.48	353.65

Applying the formula above to each feature gives these standardized scores:

Feature	Target (std.)	Candidate (std.)
Dragon Sightings	-2.6118	-1.0957
Avg. Wizard Age	+0.7595	-0.4637
Pct. Leprechaun Cottages	+0.4621	+0.4621
Goblin Family Units	-1.2455	-1.5056

Step 2 — Compute the squared difference for each feature, then sum and take the square root.

$$d = \sqrt{\sum_i (z_{\text{target},i} - z_{\text{cand},i})^2}$$

Feature	Target (std.)	Cand. (std.)	Squared difference
Dragon Sightings	-2.6118	-1.0957	2.2983
Avg. Wizard Age	+0.7595	-0.4637	1.4962
Pct. Leprechaun Cottages	+0.4621	+0.4621	0.0000
Goblin Family Units	-1.2455	-1.5056	0.0677
<i>Sum</i>			<i>3.8622</i>

$$d = \sqrt{3.8622} = \mathbf{1.9652}$$

Distance Distribution

The chart below shows the distance from the target to each of the 20 candidate rows, sorted closest to farthest. The **highlighted bar** is the selected match.

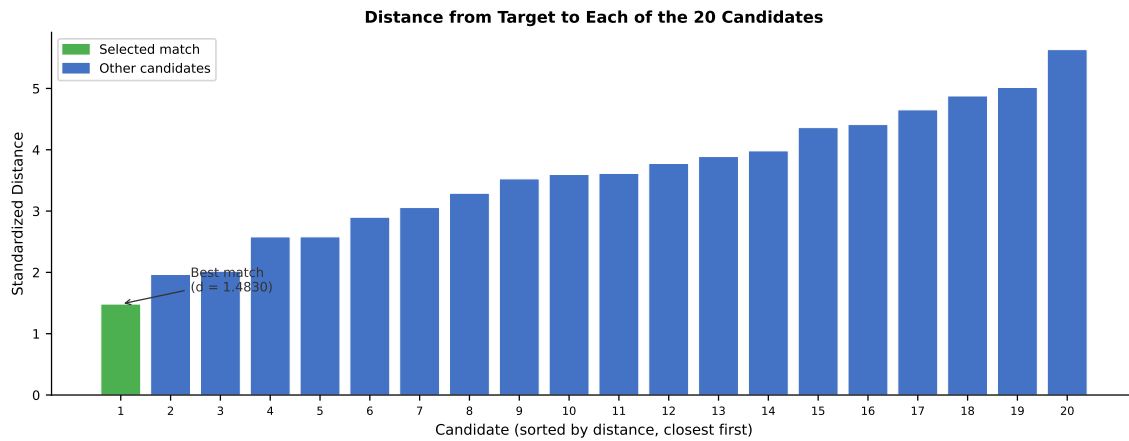


Figure 1: Distances from target to all 20 supplemental candidates, sorted ascending. The selected match is highlighted.

Signal Analysis

Each signal below is reported in the output file for every matched row. This section explains what each signal means and why it has the value it does for this particular match.

euc_distance 1.4830

The best-match distance is **1.4830** — far larger than the rounding scenario (0.2902) and orders of magnitude above the exact match (0.0000). The scale error inflates every candidate's distance by roughly the same large amount, making the absolute distance value unreliable as a quality signal on its own. The per-feature contribution table above is more informative here.

per_feature_contribution (best match)

This is the key diagnostic for this scenario. When one feature uses the wrong units, it generates an outsized squared difference that dwarfs all other features. Here **Dragon Sightings** accounts for **100.0%** of the total distance, with the remaining features contributing only 0.0%. Concentration alone is not a red flag — if only one feature differs, it will naturally account for 100% of the distance even when the difference is trivial. The warning sign here is that this concentration is paired with a large absolute distance (1.4830). Together they indicate the dominant feature is not slightly off but misscaled by orders of magnitude.

Feature	Share of distance
Dragon Sightings	100.0%
Avg. Wizard Age	0.0%
Pct. Leprechaun Cottages	0.0%
Goblin Family Units	0.0%

nndr 0.7546

$\text{NNDR} = d_1/d_2 = 1.4830 \div 1.9652 = \mathbf{0.7546}$. This is below the 0.80 threshold. Despite the large absolute distance, the best match is still meaningfully closer than the second-best — likely because the correct row's other three features match exactly, giving it a small advantage even under the scale error.

near_miss_count 0

0 supplemental row(s) fall within the near-miss threshold ($d_1/d_i \geq 0.80$). The best match is still clearly separated from the others, likely because the correct row's remaining features match the target exactly while no other row does.

mnn_confirmed True

Confirmed: True. The matched row points back to this target as its nearest record in the reverse search. The match is symmetric despite the unit error.

repeats 1

1 row(s) tied at the minimum distance. No exact tie.

smd (per feature) [0.0000, 0.0000, 0.0000, 0.0000]

With one target row, SMD is not computable and is reported as 0. In a full run, the scale error on Dragon Sightings would appear as a very large SMD on that feature — potentially the most obvious diagnostic signal of all.

flags**(none)**

Flags: (none). No flags raised. The NNDR is close to the 0.80 threshold, but the remaining three features were unique enough that the correct row still pulled ahead. The match succeeded despite the unit error.

Data note: Values in this example are adapted from U.S. Census Bureau American Community Survey (ACS) public data. Column names have been replaced with illustrative labels for demonstration purposes. This document is intended for researcher education and internal system validation.